One Recipe, Many Harnesses:
What Self-Evolution Encodes Across Languages and Models

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HarnessRSI 64 · ★★

evidence: retained

Paper v1

1 · Diagnose, edit, retain

Common minimal seed · frozen weights
Short prompt + shell; empty hooks/memory
Editable: prompts / workflow / memory / tools

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Run evolution tasks → typed diagnosis
Trajectories, patches, tests, grader, budget
Advisory routing suggests where to edit

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Propose edit + instance-level contract
Predict intended fixes and regressions
Smoke validation; at most two slots changed

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Re-run evolution split; score checkpoint
Solve rate − 2.0 × fraction of lost solves
Validate whole checkpoint, not each rule

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After three rounds → held-out test
Ship best evolution-selected checkpoint
Fallback to seed only on evolution data

Rollback is tolerant, not a strict gate: 0.12 dip tolerance; revert after two non-improving rounds or a 0.25 catastrophic drop. Seed fallback does not guarantee test non-regression.

Verification hooks nudge testing, not block submission. Main grid: policy, analyzer and driver share the model. Memory ≤12 entries × 400 characters.

Routing: A.2 and Table 2 disagree on wrong_fix.

2 · Large gains, concentrated

Selected Haiku 4.5 cells · held-out mean_solve (%)

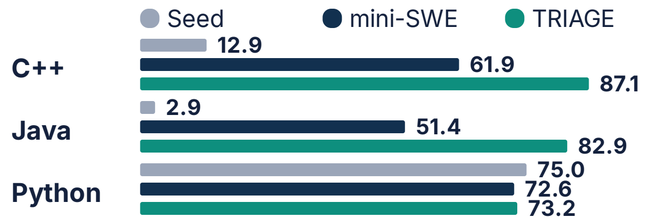


Table 3. Stated: eight languages × three models; random disjoint 20 evolution / 50 test; three rollouts, not pass-at-least-once. Table 9 totals do not reconcile; actual coverage unverifiable.

3 · Nulls and controls matter

- **GPT-5-mini:** all $|z| < 1.3$. Even C 28.3 → 36.7 is not significant; mini-SWE reaches 46.7.
- **Python:** no benefit. DeepSeek 73.2 → 66.1; paired −7.1 ± 4.2, z=−1.69 (single-star negative in Table 4).
- **DeepSeek / mini:** TS 63.5 / 65.4; Rust 35.2 / 44.4; Python 66.1 / 71.4; C tied 65.0 / 65.0.
- **Driver probe, Go/DeepSeek:** seed 18.5; GPT-OSS 34.2, DeepSeek 33.3, Sonnet 35.6. No table variance/CI.
- **Tool probe, selected task family:** 9.3 preinstalled vs 22.2 no tool, seed 11.1. Not the main C++ row.

4 · Three different kinds of reuse

Recipe: re-derive each language/model's native harness.

Transplant: unchanged source artifact; fixed DeepSeek executor; no target adaptation.

Retention = (transfer − seed) / (native − seed)

18/20 positive; 14/20 below native; mean 0.63. C++→TS 1.50 vs JS→TS −2.30. Native is not a ceiling. No pairwise variance/CI.

Universal · DeepSeek-V4-Flash only: ten all-language memory rules. JS 1.13, Go 1.08; Java 0.68, C++ 0.65, TS 0.48. Includes target evolution experience—not leave-one-language-out.

5 · Evidence and budget limits

- **Haiku Java: +80.0 ± 6.9** paired (Table 4). Rollout/resample uncertainty; one search per cell. No mini uncertainty there.
- **80 steps / $1.5 per rollout.** Static seed and released mini get no evolution/gate. No total-compute-matched baseline.
- **$256 meta + $3,667 rollouts** (Appendix I): aggregate, not per-harness matching. Donor evolution is not free.
- Disjoint language-local splits + memory-token guard; no reported cross-language overlap audit. Not evidence of leakage.
- No separate routing/contract/gate ablation. Post-hoc recovery partly reflects grader protocol.

Takeaway

A shared evolution recipe installs useful behavioral discipline around frozen models. Large held-out gains are model- and language-dependent; unchanged artifacts transfer unevenly. This is not proof of discovering new operational facts, universal dominance, or equal-budget superiority.

Our read · HarnessRSI 64 · retained

Reports: held-out tests, transfer, rollout uncertainty, driver/tool probes, optimizer + rollout costs.
Does not report: evolution-seed variance, total-compute matching, isolated gate/routing/contract effects, cross-language overlap audit.
b12 note: costs in Appendix I; variance "no" means no repeated searches.